

Brazilian E-Commerce Public Dataset by Olist

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Abstract. *The rapid growth of e-commerce has increased the importance of understanding customer satisfaction and logistics performance to support data driven business decisions. This study utilizes the Brazilian E-Commerce Public Dataset by Olist, there are 100,000 transactions involving customers, products, payments, reviews, and delivery information. This research combines exploratory and data analysis and machine learning techniques to identify factors influencing customer satisfaction and evaluate logistics performance. The features are delivery time, estimated delivery time, and delivery delay. Random Forest Regression Model was developed to predict customer review score using transaction and logistics variables. Customers who received their order faster will give a higher review score, but while delayed the review will get negative ratings. Feature importance analysis is delivery time and delivery delay were the most influential predictors of customer reviews.*

Keywords: *E-Commerce, Customer Satisfaction, Random Forest Regression, Logistics Performance, Delivery Time, Machine Learning, Data Analytics, Olist Dataset*

I. INTRODUCTION

The rapid growth of digital technology and internet adoption has transformed the global retail industry, making e-commerce the fastest growing markets in Latin America, supported by a growing middle class, widespread mobile usage, and increasing online shopping activities. This growth has intensified competition among e-commerce platforms and highlighted the importance of data driven business strategies [12],[22]. The growth is also supported by the rapid adoption of digital payment technologies. These innovations have increased transaction and contributed to the expansion of online shopping activities across various consumer segments [5], [21].

Customer satisfaction is essential for the long-term success of e-commerce platforms as it directly impacts on customer loyalty, reviews, and brand reputation. One of the many operational factors that

have been identified as being among the strongest determinants of customer satisfaction is delivery performance. Customer satisfaction prediction is an important analytical task because of the negative reviews and less trust in online platforms due to delays in delivery [1], [2], [4].

In recent years, machine learning has made it possible for companies to forecast customer satisfaction with their operations and transactions more accurately. Random Forest, XGBoost, Support Vector Machine and Decision Tree are some algorithms that have performed well in predicting. Studies also show the feature engineering and appropriate model selection to improve prediction accuracy [14],[15]. The amount of e-commerce transactions has increased, and with this, machine learning and data analytics have gained significant importance in understanding

customer behavior, optimizing performance, and aiding in business decisions [10, 8].

Customer segmentation is also a key factor in creating successful marketing strategies, besides satisfaction prediction. The RFM analysis was used to identify customer groups with similar purchasing behavior using clustering techniques such as K-Means and DBSCAN. This knowledge helps businesses create more targeted marketing strategies and enhance customer retention [23],[24].

Another important one is logistics performance, as Delivery delays can lead to higher operational expenses, lower customer satisfaction, and lower seller performance. Product characteristic, geographic distance and location of the seller are some of the factors that cause delay in shipment which can be used to improve the logistics efficiency and service quality of the company [2],[4].

More than 100,000 orders are used in this study, and the dataset includes information on customer, sellers, products, payment, review, and delivery performance, all from the Brazilian E-Commerce public dataset provided by Olist. The richness of the data allows it to use it for customer satisfaction prediction, customer segments and logistics performance analysis [1],[16].

The objective of this research is to perform a thorough analysis with the help of machine learning and data analytics techniques. This study aims to achieve three goals: (1) predicting customer satisfaction using ML classification models, (2) developing customer segmentation with the help of RFM based clustering and (3) analysing shipping delays and their effect on customer satisfaction. It is believed that these are expected to give insight into the improvement of customer experience, marketing strategies and logistic performance [15].

II. LITERATURE REVIEW

2.1 E-Commerce Growth in Brazil

Brazil is the largest e-commerce market in Latin America and one of the fastest growing digital economies in the world. In 2023 online retail sales amounted to about BRL 186 billion and are

projected to increase at a 18,91% CAGR through 2030 [19],[23]. The rise is attributed to the rising penetration of the Internet, the use of smartphones and the trust of consumers in digital transactions. Platforms such as Olist serve the ecosystem by providing a connection point for SMEs to various marketplaces and provide valuable opportunities for e-commerce analytics research [16],[20]. Digital payment solutions have also been embraced, further driving e-commerce growth. The market reports in recent times suggest that real time payment systems and alternative payment methods are emerging as key enablers of digital commerce in the Latin American region [7, 13, 18].

2.2 Customer Satisfaction and Delivery Service Quality

Customer satisfaction is an important element of e-commerce success and logistics service quality plays an important role in customer satisfaction. The time of delivery, tracking and security of the product play a major role in customer satisfaction and loyalty [4]. Last mile delivery efficiency is crucial in customer retention and delivery delays decrease trust and customer experience [2]. Optimization of delivery is still a significant concern for e-commerce platforms because of its high cost of last mile logistics [17]. Secure and convenient payment experiences are also important for customer trust and satisfaction [25].

2.3 Machine Learning Based Customer Satisfaction Prediction

Machine learning has become widely used for predicting customer satisfaction in e-commerce. Research using Olist dataset shows that features like delivery duration and product rating improve predictive performance [14]. Comparisons between traditional ML models and deep learning indicate that more complex models do not provide better results, the importance of feature selection and model suitability [15]. Delivery time, order value, and product category have consistently been identified as important predictors of customer satisfaction [14], [15]. Recent studies show that machine learning techniques can effectively identify behavioral patterns and support prediction analytics, including customer behavior analysis, sales forecasting, and satisfaction prediction [10], [8], [6].

2.4 Customer Segmentation and RFM Analysis

Customer segmentation is essential for developing effective marketing strategies. The RFM model is commonly used to classify customers based on purchasing behavior. Previous studies combined RFM analysis with clustering like K-Means and Hierarchical Clustering to identify high value [24]. The RFMT model is used to improve time dimension and apply multiple clustering algorithms with validation metrics to determine optimal customer groups [23]. ML based RFM supports personalized marketing and customer retention strategies [9].

2.5 Sentiment Analysis of Customer Reviews

Customer reviews provide valuable information about consumer experiences and perceptions. NLP combined with machine learning algorithms like Naive Bayes, SVM, and Random Forest show strong performance in sentiment classification tasks [11]. The Olist dataset, reviews complement numerical rating and improve customer satisfaction prediction by identifying dissatisfaction signals more effectively [1], [11].

2.6 Customer Churn Prediction and Retention

Customer churn is a major challenge because retaining existing customers is generally more cost effective than acquiring new ones. Machine learning models have been successfully used to identify churn risk using factors such as customer complaints, delivery distance, and customer tenure [20]. Customer segmentation improves retention strategies by enabling businesses to target customers who are more likely [9], [19]. Predictive analytics enables businesses to identify customers with higher probability of churn and develop personalized retention strategies based on purchasing behavior and transaction history [3], [25].

2.7 Brazilian E-Commerce Public Dataset (Olist)

This study uses the Brazilian E-Commerce Public dataset released by Olist through Kaggle [16]. The dataset contains more than 100,000 orders and includes information about customers, sellers, products, payments, reviews, and geolocation data [1]. The comprehensive features make it suitable for customer satisfaction prediction, customer

segmentation, consumer behavior analysis, and logistics performance evaluation [14], [15], [16], [17], [20], [23], [24].

III. METHODOLOGY

This study utilizes the Brazilian E-Commerce Public Dataset by Olist, a comprehensive transactional dataset that records customer orders, product details, payment information, logistics data, and customer reviews. The dataset used in this analysis has been pre-cleaned and stored as `final_dataset_clean.xlsx`. The analysis was conducted using Python, leveraging core data science libraries including Pandas and NumPy for data manipulation, and Matplotlib and Seaborn for data visualization.

3.1 Data Preprocessing and Feature Engineering

The data were subjected to a number of pre-processing steps prior to analysis. All date related columns such as `order_purchase_timestamp`, `order_approved_at`, `order_delivered_carrier_date`, `order_delivered_customer_date`, `order_estimated_delivery_date` and `shipping_limit_date` were changed to proper datetime objects. Through this conversion, time based feature extraction and trend analysis could be performed.

After the conversion, some new features were designed to enhance the data set:

- The order timestamp was split into `purchase_month`, `purchase_day`, and `purchase_hour` to reflect the temporal purchasing patterns.
- The `total_cost` was calculated by adding price and `freight_value` to get the total cost paid per item.
- The time taken for delivery was determined as the number of days from the purchase date to the actual delivery date, and recorded as `delivery_time`.
- The difference between estimated delivery date and purchase date was used as `estimated_delivery_time`.
- The actual delivery time minus the estimated delivery time was calculated as `delivery_delay` and indicates the order's delivery time being earlier or later than expected.

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted across multiple dimensions to understand customer behavior, logistics performance, and product-level patterns. The following aspects were examined:

- Distribution of review scores across all transactions.
- Distribution and outlier detection of delivery times.
- Relationship between delivery time and customer review scores.
- Payment method usage and its association with review scores.
- Top 10 most-purchased product categories and their respective review score distributions.
- Monthly revenue trends to identify seasonal purchasing behavior.
- Freight value distribution and its potential influence on customer satisfaction.
- Geographic distribution of customers and delivery time variation by state.
- A correlation heatmap across all key numerical variables.

3.3 Predictive Modelling

A Random Forest Regression model was employed to predict the customer review score (review_score) based on transaction and logistics features. The features used as predictors were: price, freight_value, payment_installments, payment_value, product_weight_g, product_length_cm, product_height_cm, product_width_cm, delivery_time, and delivery_delay.

The data set was divided into 80% training and 20% testing data with a fixed random state for repeatability. 100 decision trees were used for training the model (n_estimators=100). Three performance metrics were used to assess model performance: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2 Score). The scores of feature importance were also extracted, which are the most influential variables for predicting customer satisfaction.

	Model	MAE	RMSE	R2
1	XGBoost	0.976387	1.268846	0.134002
0	Random Forest	1.008904	1.289975	0.104920

We also used a second model, XGBoost, to validate the results obtained from the first model. The evaluation showed that both models produced similar performance metrics and predictions. This consistency indicates that the results are reliable and suggests that the first model is valid for this analysis.

IV. RESULT & DISCUSSION

4.1 Review Score Distribution

The scores of the customer reviews are highly skewed to the right, with most customers giving the highest score of 5. Overall, the pattern indicates that the satisfaction level of customers using the Olist platform is fairly high. But a significant share of the customers who were rated low, especially 1, suggesting a portion of users had very poor shopping experiences. This bimodal distribution, with reviews at both ends, is a widespread phenomenon in e-commerce rating systems, and deserves further investigation into the root causes of dissatisfaction.

4.2 Delivery Time Analysis

The distribution of delivery times indicates that the majority of orders were delivered within a reasonable time. The histogram, with a kernel density estimate (KDE) overlaid, shows that the distribution is right-skewed, with most deliveries falling within the lower range of time. There are, however, some interesting exceptions, or outliers, corresponding to orders with very long delivery times. These extreme cases are probably responsible for the poor customer review scores.

This is also reflected in the box plot showing the delivery time for each group of reviews by score. The average delivery time for customers who rated the review score 1 was significantly longer than the average delivery time for customers who rated the review score 4 or 5. As review scores drop, the interquartile ranges get wider and the median scores get higher. The Pearson correlation between delivery_time and review_score is negative, meaning that statistically there is a negative

correlation between delivery time and customer satisfaction.

4.3 Payment Method Distribution and Review Scores

The most popular payment method among Olist customers is credit card, followed by boleto, which is a payment slip system in use in Brazil, and to a lesser extent, debit cards and vouchers. This discovery is aligned with the overall trend of consumers in Brazil's e-commerce landscape who favor flexible and credit-based payment methods.

Boxplots of the distributions of the review scores by payment method show similar distributions. The median review scores are similar for each payment method, and there is no significantly different satisfaction result for any payment method. This means that the type of payment is not a significant factor in customer satisfaction and dissatisfaction is more likely to be due to operational or logistics issues.

4.4 Product Category Analysis

The top 10 most ordered product categories are displayed in a bar chart, revealing clear preferences among consumers on the platform. Some product categories like home goods, health and beauty, and sports and leisure are always among the top transactional volume categories and show a high level of market demand. These are high volume categories, probably the revenue streams that make up the bulk of the platform's revenue.

The boxplot of review scores by product category shows that there is meaningful variation in customer satisfaction by product category. There are categories with more concentrated distributions of high scores, and others with more spread distributions of low score outliers. This variation may be due to varying quality of products, descriptions of items, packaging standards, or delivery difficulties with items in the category. Those categories that have consistently lower satisfaction scores should be targeted at quality audits and seller performance reviews.

4.5 Monthly Revenue Trends

The line chart of monthly revenue (payment_value) shows fluctuations throughout the year, with certain

months recording notably higher total payments. These peaks likely correspond to promotional events, holidays, or seasonal shopping periods that drive increased purchasing activity. The presence of such patterns suggests opportunities for targeted marketing campaigns and inventory planning around high-demand periods.

4.6 Freight Value and Customer Satisfaction

The distribution of freight values is strongly skewed to the right as most transactions have relatively low freight costs. But some transactions have high freight rates, probably due to distance, size of product or weight.

Freight value vs. review score has a slight negative trend with customers with higher freight values giving lower ratings. This can be a result of customers not being satisfied with the value of the shipping cost, especially if it is high, and the delivery time is longer or delayed.

4.7 Geographic Analysis

As seen in the top customer states chart, the number of transactions is concentrated in a few states, and follows the pattern of population density and economic development in the regions of Brazil. The states of São Paulo and Rio de Janeiro are the most well-represented, as they are major economic centers.

The box plot of delivery time by state shows significant differences between the states. The delivery time becomes longer as the distance from the major distribution centers increases, and the interquartile range is larger, indicating that there is more variation. This regional delivery gap is an important operational issue, as it impacts customers in less-served areas and is likely to lead to lower satisfaction scores in less-served areas.

4.8 Correlation Analysis

There are several interesting relationships between numerical variables shown in the correlation heatmap. The most positive correlation is between payment_value and price, as would be expected as the amount of payment is directly related to the price. Product weight and dimensions (length, height, width) also have moderate positive correlations with each other, indicating a physical

relationship between the product dimensions and the product weight.

Interestingly, there is a negative correlation between `delivery_time` and `review_score`. This correlation is not large, but has the same direction as all previous visualisations and further confirms the main result that faster deliveries are correlated with higher customer satisfaction ratings in a systematic way.

4.9 Predictive Modelling Results

The ten features selected were used to train the Random Forest Regression model for predicting the customer review scores. The metrics of the evaluation demonstrate that the model has a fairly good predictive performance. The MAE and RMSE values are considered acceptable because the scores of the reviews are subjective and subjective factors which are not included in the data affect them.

The `delivery_time` and `delivery_delay` are the two most important features for the `review_score`, significantly more important than the others. The result supports the EDA findings, and validates that logistics performance is indeed the most important operational aspect of customer satisfaction on Olist. Other moderately important features are `payment_value` and `price`, indicating a degree of effect of transaction size on customer satisfaction.

The scatterplot of actual vs. predicted review scores reveals that the model concentrates around the middle of the score distribution, and the model tends to capture the general direction of the high and low review scores. Ensemble regression models on ordinal outcomes tend to have more deviations at the extremes, especially at very low scores.

In this study, Olist's Brazilian E-Commerce Public Dataset was analyzed in depth, using exploratory data analysis and predictive modelling to gain insights into the factors that influence customer satisfaction. A consistent and compelling observation is that the performance of the delivery is the most important one that determines the customer's rating of the product or service, across all analytical approaches, visualizations, correlation analysis and machine learning.

Customers who get their orders sooner and nearer to the promised delivery date always rate their orders higher. Delays, however, in absolute delivery time and in comparison to estimates, are strongly related to poor ratings, on the other hand. This relationship is true for geographic regions, product categories and payment methods, indicating that this is a systemic phenomenon and not a result of isolated cases.

Secondary findings suggest that there is a moderate effect of product category on satisfaction, with some categories having consistently lower review scores, which may be because of quality or packaging issues. There are also significant operational issues related to geographic differences in delivery time between states in Brazil, especially for customers in less connected areas. While not the main factor, there is a slight negative relationship between satisfaction and freight costs if freight costs are high.

V. CONCLUSION

Based on these findings, three strategic recommendations are proposed for platform operators and sellers. First, logistics optimization should be the highest priority, reducing delivery times and improving the accuracy of delivery estimates will have the most direct and measurable impact on customer satisfaction scores. Second, product quality assurance efforts should be directed toward underperforming categories, incorporating improved packaging standards and clearer product descriptions. Third, regional logistics infrastructure, particularly warehouse placement and last-mile delivery partnerships, should be strengthened in states with consistently high delivery times.

It is important to acknowledge the limitations of this analysis. The dataset does not capture external factors such as product quality perception, customer expectations, weather disruptions, or regional infrastructure quality, all of which may independently influence satisfaction. Furthermore, the relationships identified in this study are correlational in nature and do not establish direct causality. The Random Forest model, while informative, may also exhibit different performance characteristics when applied to live or out-of-sample data.

Future research could benefit from incorporating customer demographic data, seller performance metrics, and natural language processing of written review comments to capture the qualitative dimensions of satisfaction that numerical scores alone cannot fully represent. Additionally, applying classification-based models or deep learning approaches may further improve predictive accuracy for this inherently ordinal outcome variable.

VI. REFERENCES & ACKNOWLEDGEMENT

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Link Coding Analysis:

https://colab.research.google.com/drive/1wcqsNgSp6-EK7s_JN4ORL6Y4aT7sMfOV?usp=sharing

Link to Dataset :

https://www.kaggle.com/datasets/olistbr/brazilian-e-commerce/data?select=olist_products_dataset.csv